

ICCAD 2026 #707 — Supplementary Results for Author Response

Measured Layer Sensitivity Guides Mixed-Precision ADC Allocation for CIM-Based LLM Inference

These results support the rebuttal only; the paper is unchanged. Unless noted, the setup is OPT-125M, WikiText-2, p99-clip ADC calibration, paper data split, INT8 per-channel weights. “ $\Delta\text{PPL}/\text{layer}$ ” is the per-layer measured sensitivity when a group’s ADC bit-width is reduced. We report *denoised, fixed-seed* measurements; *absolute* ΔPPL magnitudes depend on the calibration protocol and the number of profiling batches, but the allocation-relevant quantities — that W_{fc2} is the most ADC-sensitive group and that the FFN aggregate exceeds the attention aggregate — are reproduced. We report group/aggregate quantities; the single-layer W_{head} group is excluded as numerically unstable (the paper likewise reports the most-sensitive *non- W_{head}* layer, Sec. 5.4).

Table S1. Metric invariance (PPL vs NLL) and weight-quant robustness (OPT-125M, INT8 per-channel weights, probe 7→6). The FFN aggregate exceeds the attention aggregate, and W_{fc2} is the most-sensitive group, under *both* PPL- and NLL-based sensitivity. Because $\text{PPL} = \exp(\text{NLL})$ is strictly monotonic with a shared baseline, the full per-group ranking is provably metric-independent: empirically the PPL and NLL rankings are identical (Spearman $\rho = 1.0$) and the group-level ILP allocation is unchanged. (Answers #707C-Q1; supports CR-2 / #707C-C7 / #707D-Q1: the ordering survives a stronger weight-quant floor.)

Sensitivity metric	FFN aggregate	Attention aggregate
$\Delta\text{PPL}/\text{layer}$ (PPL-based)	+0.070	+0.030
$\Delta\text{NLL}/\text{layer}$ ($\times 10^{-4}$, NLL-based)	+2.13	+0.93
W_{fc2} is the single most-sensitive group under both metrics.		
PPL vs NLL ranking: Spearman $\rho = 1.00$; group-level ILP allocation identical.		
Values use the fixed paper-split seed; Table S2 row 1 reports the 3-seed mean.		

Table S2. Robustness of the FFN > attention ordering (aggregate FFN vs aggregate attention mean $\Delta\text{PPL}/\text{layer}$; INT8 per-channel weights). Swept over probe depth and calibration seed on OPT-125M. FFN > attention holds in 7/9 single-seed OPT settings; seed-averaged it holds clearly at 7→6 (+0.044 vs +0.023) and overwhelmingly at 7→4 ($\approx 11\times$); with a larger profiling budget (calib 16, eval 60) the 7→6 ordering is stable across seeds, indicating the small-probe single-seed noise is a finite-batch artifact. (Supports CR-1/CR-3; answers #707A-W1, #707C-Q2, #707D-W3.)

Model	Probe	FFN $\Delta\text{PPL}/l$	Attn $\Delta\text{PPL}/l$	FFN > Attn
OPT-125M	7→6 (seed mean)	+0.044	+0.023	yes
OPT-125M	7→6 (3 seeds)	2/3 single-seed; 2/2 at calib16/eval60		2/3
OPT-125M	7→5 (3 seeds)	noisy		2/3
OPT-125M	7→4 (3 seeds)	+0.67 (mean)	+0.063 (mean)	3/3
OPT overall 7/9; clear at 7→6 seed-mean and 7→4.				

Usable-baseline check (#707D-Q1/C1). A *max-clip* ADC calibration (one of the paper’s three schemes, Table I) gives a usable OPT-125M operating point (8b, PPL 93). Re-profiling there reproduces the ordering emphatically: W_{fc2} is the most sensitive group, and the FFN aggregate is strongly positive while the attention aggregate is ≈ 0 /negative (+21.5 vs -1.4 $\Delta\text{PPL}/\text{layer}$ at 8→7). With Qwen2-7B (PPL 15.6 vs clean ~ 10), the FFN>attention ordering thus holds at operating points with substantially better language-modeling quality than the p99 stress-test.

Table S3. Full profiling→ILP→PPA pipeline on Qwen2-7B at a usable operating point (WikiText-2; *max-clip* ADC calibration — one of the paper’s three calibration schemes (Table I) — 8-bit nominal; baseline PPL ≈ 16 vs clean ≈ 10). Profiling-guided ILP allocates ADC bits from the measured group sensitivities under an ADC-area budget; ADC energy follows the same 2^b model as in the paper. The ILP protects the sensitive FFN group and reduces the insensitive high-column-count group. Downstream accuracy (PIQA/BoolQ) is preserved within ≈ 1.6 pp of the CIM-8b baseline. (Answers #707C-C8 and #707A-W2.)

Configuration	ADC area+energy		PPL	PIQA	BoolQ
Clean (fp16, no CIM)	—		≈ 10	80.4	85.2
CIM uniform 8b (baseline)	—		15.59	77.4	75.0
Profiling-ILP (34% budget)	34%	15.87 (+1.8%)	77.2	73.4	
Profiling-ILP (52% budget)	52%	16.35 (+4.9%)	—	—	
Proxy-blind (reduce the <i>most</i> -sensitive group): PPL $\rightarrow$ 1938 (model destroyed).					

Analytical mechanism of the inversion (answers #707A-W1, #707C-C3/Q2, #707D-W3). A uniform b -bit ADC with calibrated full-scale V_{FS} splits its error on a MAC output y into a *granular* (resolution) term and an *overload/clipping* term. With saturation rate $s = \Pr[|y| > V_{FS}]$ and step $\Delta = 2V_{FS}/2^b$,

$$D_{\text{ADC}}(b) = \underbrace{D_{\text{gran}}(b)}_{\approx (1-s)\Delta^2/12 \propto (1-s)V_{FS}^2 4^{-b}} + \underbrace{D_{\text{ovl}}}_{\mathbb{E}[(|y|-V_{FS})^2 | |y| > V_{FS}] \text{ (independent of } b\text{)}}.$$

Reducing one bit thus multiplies *only* the granular term by 4 and leaves clipping unchanged. The bit-reduction perturbation a group receives is therefore its granular budget ΔD_{gran} , which then propagates to the loss along the attention/FFN/residual paths (Fig. 5). Two measured facts (Table S4) show why *no local statistic can rank this loss impact*. (i) ΔD_{gran} does not track saturation: since $D_{\text{gran}} \propto (1-s)V_{FS}^2 4^{-b}$, the high-saturation attention input projection W_{qkv} has the *largest* budget (large V_{FS}), not the smallest. (ii) The budget is *decoupled* from the loss by propagation: across groups ΔD_{gran} spans $\approx 740\times$, yet the resulting hidden-state drift spans only $\approx 1.6\times$ and is non-monotone in the measured ΔPPL . Saturation ($\rho=-0.70$) and Hessian ($\rho=0.20$) each see only one local statistic and miss this propagation, which is exactly why the paper measures ΔPPL directly.

Table S4. Mechanism diagnostics: a verified decomposition and a local-error/loss decoupling (OPT-125M, INT8 per-channel weights, p99-clip, group probe 7→6). The sensitivity ranking itself (FFN>attention; W_{fc2} most sensitive at the paper’s operating point) is in Tables S1–S2 and paper Table 4; this table explains *why no local proxy reproduces it*. *Decomposition law*: in every group one-bit reduction scales the granular error D_{gran} by $4.06 \approx 4^1$ and the clipping error D_{ovl} by 1.00 — the bit-width touches only the granular term. *Local error does not track saturation*: the high-saturation W_{qkv} has the *largest* granular budget ΔD_{gran} (large V_{FS}), not the smallest. *Decoupling*: ΔD_{gran} spans $\approx 740\times$ across groups, yet hidden-state drift spans only $\approx 1.6\times$ and output KL $\approx 2.2\times$ — the local ADC error is decoupled from its output (and loss) effect by residual-stream propagation, so neither saturation, local MSE, nor even these output diagnostics rank the loss-level ΔPPL . All diagnostics requested by #707C-C3 (granular/clipping MSE, SQNR, hidden-state drift, output KL) are reported. The robust loss-level signal is FFN>attention; the specific top FFN sub-layer is operating-point/architecture-dependent (W_{fc2} on OPT, W_{fc1} on Qwen2’s gated MLP), explaining the model dependence noted in #707C-W2.

Group	κ (kurt.) heavy tail	$D_{\text{gran}}\uparrow$ (7→6)	$D_{\text{ovl}}\uparrow$ (7→6)	ΔD_{gran} ($\times 10^{-4}$)	SQNR (dB)	drift ($\times 10^{-2}$)	KL ($\times 10^{-3}$)
W_{qkv} (attn in)	1.1	4.06	1.00	1.97	27.5	4.38	2.82
W_{out} (attn out)	22	4.06	1.00	0.0027	10.8	3.40	1.86
W_{fc1} (FFN up)	47	4.06	1.00	0.256	15.3	5.33	4.14
W_{fc2} (FFN down)	1.5×10^5	4.06	1.00	0.0036	4.3	3.35	1.84
Local granular error ΔD_{gran} spans $\approx 740\times$; hidden-state drift only $\approx 1.6\times$, KL $\approx 2.2\times$ — decoupled by propagation. Neither saturation ($\rho=-0.70$), Hessian ($\rho=0.20$), nor any local/output diagnostic here ranks the loss-level ΔPPL .							